

S.No	ESP32 Notation	Signal Type	Signal Description	Purpose	Interlinked Signal(s)	Interlink Logic (If / Then)	Test Value to Inject (Rig)	Expected Response	Monitoring Method	Carel Modbus Counterpart
1	DO-1	DI	Auto/Manual Status	Monitoring	EC Fan's Speed (AI-3); DPT (AO-4)	Auto: EC Fan Speed follows DPT PID loop (Sec 2.1). Manual: fan runs from local potentiometer, ignores DPT (Sec 2.2).	Toggle Auto/Manual while varying DPT (AO-4)	In Auto, EC Fan Speed (AI-3) tracks DPT changes. In Manual, EC Fan Speed stays constant regardless of DPT.	Controller display + BMS (Modbus)	Discrete Input 0 AUTO/MANUAL STATUS
2	DO-2	DI	Filter Status	Monitoring	None (standalone alarm)	When filter DP exceeds client-set threshold, Filter Dirty Alarm raised on display & BMS (Sec 9).	Toggle contact to simulate DP switch trip	Filter Dirty Alarm appears on controller display and is readable via BMS.	Controller display + BMS alarm bit	Discrete Input 1 PRE FILTER STATUS or 2 FINE FILTER STATUS
3	DO-3	DI	Fire Status	Interlock	AHU On/Off Command (DI-1); EC Fan's Speed (AI-3); CHW/HW Valve Control (AI-1/AI-2)	NC in normal condition; acts as NO in fire → AHU power shut down (Sec 10). Soft alarm raised on display & BMS.	Open the Fire Status contact (simulate fire = NO)	AHU On/Off Command goes OFF, EC Fan Speed drops to 0%, soft fire alarm active on display/BMS.	Controller display + BMS soft alarm	Discrete Input 4 FIRE STATUS
4	DO-4	DI	EC Fan Trip status	Interlock	EC Fan's Speed (AI-3); AHU status (DO-5)	MPCB NC contact opens on trip → controller stops commanding fan, raises alarm (Sec 6).	Open MPCB contact simulation (trip)	EC Fan Speed command drops to 0 / fan stop commanded; alarm on display & BMS.	Controller display + BMS	Discrete Input 5 EC FAN TRIP STATUS
5	DO-5	DI	AHU status	Interlock	AHU On/Off Command (DI-1)	Reflects actual contactor state. Sec 11 lists "AHU Offline" as a BMS alarm	Toggle independently of AHU On/Off Command to simulate a contactor-feedback mismatch	If commanded ON but status shows OFF, expect an "AHU Offline" alarm	Controller display + BMS ("AHU Offline" alarm bit)	
6	DI-1	DO	AHU On/Off Command	Monitoring & Command	Fire Status (DO-3); EC Fan Trip (DO-4); AHU status (DO-5)	Enabled by BMS/local command; controller checks fan fault, fire signal, door status before starting (Sec 1). Forced OFF by active Fire or Fan-trip interlock regardless of command.	Send ON/OFF via Carel Coil 0, or trigger an interlock (Fire/Trip)	DI-1 reading follows the command	Controller display + BMS	
7	DI-2	DO	UV Lamp On/Off Command	Monitoring & Command		Direct ON/OFF via Relay-3, BMS-writable (Sec 7).	Command Relay-3 ON/OFF via BMS write	DI-2 feedback reflects the commanded state.	Controller display + BMS	
8	DI-3	DO	Damper On/Off Command	Monitoring & Command		Direct ON/OFF via Relay-4, BMS-writable (Sec 7).	Command Relay-4 via BMS	DI-3 feedback reflects the commanded state.	Controller display + BMS	
9	AO-1	AI	Return Air Temperature	Monitoring & Command	CHW Valve Control (AI-1)	RAT > 25°C setpoint → CW Valve modulates OPEN. RAT < 25°C → CW Valve modulates CLOSE (Sec 3).	Inject 30°C (above setpoint), then 20°C (below setpoint)	At 30°C, CHW Valve Control (AI-1) increases/opens. At 20°C, it decreases/closes.	Controller display + BMS; cross-check CW Valve FBK (AO-5)	Input Register 1 RETURN AIR TEMP
10	AO-2	AI	Return Air Humidity	Monitoring & Command	HW Valve Control (AI-2)	RAH > 50% setpoint → HW Valve modulates OPEN. RAH < 50% → HW Valve modulates CLOSE (Sec 4).	Inject 70% (above setpoint), then 30% (below setpoint)	At 70%, HW Valve Control (AI-2) opens/increases. At 30%, it closes/decreases.	Controller display + BMS; cross-check HW Valve FBK (AO-6)	Input Register 2 RETURN AIR HUMI
11	AO-3	AI	Supply Air Temperature	Monitoring			Inject any value	Value displays correctly on controller and is readable via BMS; no secondary control action expected.	Controller display + BMS read	
12	AO-4	AI	Differential Pressure Transmitter (DPT)	Monitoring & Command	EC Fan's Speed (AI-3); gated by Auto/Manual Status (DO-1)	In Auto mode only: DPT > 400 Pa → fan ramps down (PID). DPT < 400 Pa → fan ramps up (PID) (Sec 2.1).	With Auto/Manual = Auto, inject 600 Pa then 200 Pa	At 600 Pa, EC Fan Speed (AI-3) ramps down over time. At 200 Pa, it ramps up.	Controller display + BMS; watch EC Fan Speed trend for PID response	Input Register 0 DPT
13	AO-5	AI	CW Valve FBK	Monitoring	CHW Valve Control (AI-1)	independent actuator feedback for diagnostic comparison, not a closed-loop control input.	Inject a diverging FBK value	No direct control action expected; diagnostic comparison only.	Controller display + BMS; compare against CHW Valve Control (AI-1) for actuator health	Input Register 4 CW VALVE OUTPUT % (Medium)
14	AO-6	AI	HW Valve FBK	Monitoring	HW Valve Control (AI-2)	independent actuator feedback for diagnostic comparison, not a closed-loop control input.	Inject a diverging FBK value	No direct control action expected; diagnostic comparison only.	Controller display + BMS; compare against HW Valve Control (AI-2)	
15	AI-1	AO	CHW Valve Control	Monitoring & Command	Return Air Temperature (AO-1); CW Valve FBK (AO-5)	Driven by RAT vs. 25°C setpoint (Sec 3)	Vary AO-1 (RAT) and observe this output	Tracks RAT per Sec 3 logic.	Controller display + BMS; rig reads this directly as AI-1	Input Register 4 CW VALVE OUTPUT % (High)

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16	AI-2	AO	HW Valve Control	Monitoring & Command	Return Air Humidity (AO-2); HW Valve FBK (AO-6)	Driven by RAH vs. 50% setpoint (Sec 4)	Vary AO-2 (RAH) and observe this output	Tracks RAH per Sec 4 logic.	Controller display + BMS; rig reads this directly as AI-2	
17	AI-3	AO	EC Fan's Speed	Monitoring & Command	DPT (AO-4); Auto/Manual Status (DO-1); Fire Status (DO-3); EC Fan Trip (DO-4)	Auto: follows DPT PID loop (Sec 2.1). Manual: driven by local potentiometer, not this signal (Sec 2.2). Forced to 0/stopped on Fire trip or Fan-trip interlock (Sec 6, 10).	Vary DPT in Auto mode; separately trigger Fire/Fan-trip interlocks	Tracks DPT per PID in Auto; drops to 0 immediately on Fire/Trip regardless of DPT.	Controller display + BMS; rig reads this directly as AI-3	Input Register 3 FAN OUTPUT %
18	SI	SI	EC Fan's Data	Monitoring	EC Fan's Speed (AI-3)		N/A — read-only diagnostic telemetry, not independently injectable			